2.3 Group 17

Question Paper

Course	CIE A Level Chemistry
Section	2. Inorganic Chemistry
Topic	2.3 Group 17
Difficulty	Easy

Time allowed: 30

Score: /21

Percentage: /100

Question 1a

Explain why fluorine has a lower boiling point than chlorine.

[2 marks]

Question 1b

Chlorine can be reacted with sodium hydroxide to form bleach.

Write an ionic equation for the reaction between chlorine and cold dilute sodium hydroxide solution.

[1 mark]

Question 1c

Give the oxidation numbers of chlorine in each of the chlorine containing ions formed in the reaction in part **(b)**.

[1 mark]

Question 2a

Halogens react with hydrogen gas forming hydrogen halides.

i)

Write the equation for the reaction of bromine with hydrogen. State symbols are not required.

[1]

ii)

State the trend in thermal stability of the hydrogen halides down Group 17. Explain your answer.

Trend ....

Explanation ....

[4]

[5 marks]

Question 2b

In the reaction outlined in part **(a)**, bromine, Br_2, is acting as an oxidising agent.

Order the halogens below from the weakest to strongest oxidising agent.

Bromine, Br_2

Chlorine, Cl_2

Iodine, I_2

[1 mark]

Question 2c

Explain why an iodide ion, I^-, is a better reducing agent than a bromide ion, Br^-.

[3 marks]

Question 3a

Write an equation, including state symbols, to show the reaction between silver nitrate solution, $AgNO_3$ (aq), and sodium iodide solution, NaI (aq).

[2 marks]

Question 3b

Sodium iodide reacts with concentrated sulfuric acid via the following equation.

$$H_2SO_4 (l) + NaI (s) \rightarrow HI (g) + NaHSO_4 (s)$$

The hydrogen iodide gas, HI (g), formed can react in a number of ways.

i)
The hydrogen iodide formed reacts with concentrated H_2SO_4 to form a yellow solid, a smell of bad eggs and a purple vapour. Name the three products responsible for these observations.
yellow solid ....

smell of bad eggs ....

purple vapour ....

[3]

iii)
Write the equation which forms the yellow solid and purple vapour from HI and H_2SO_4. Include state symbols in your answer.

[2]

[5 marks]

Question 3c

When sodium bromide reacts with concentrated sulfuric acid, bromine and sulfur dioxide are formed.

Write the half equation for the formation of bromine from bromide ions.

[1 mark]

Model Answers: Easy

1a

a) Fluorine has a lower boiling point than chlorine as:

- Fluorine is smaller than chlorine; [1 mark]
- The force between fluorine/ F_2 molecules are weaker (than the forces between chlorine/ Cl_2; molecules); [1 mark]

[Total: 2 marks]

- Make sure you use comparative words when asked to compare two substances
  - e.g. smaller/ larger
- The forces here are between fluorine and chlorine **molecules,** not atoms
  - **Tip:** Be specific with your language as examiners will deduct marks

1b

b) The ionic equation for the reaction between chlorine and cold dilute sodium hydroxide solution is:

- $Cl_2 + 2OH^- \rightarrow OCl^- + Cl^- + H_2O$; [1 mark]

[Total: 1 mark]

- The full equation for the reaction between chlorine and cold dilute sodium hydroxide is:
  - $Cl_2\ (aq) + 2Na^+ + 2OH^- \rightarrow NaCl\ (aq) + NaClO\ (aq) + H_2O\ (l)$
- We can find the ionic equation by splitting any ionic substances up and removing spectator ions
  - $Cl_2\ (aq) + \cancel{2Na^+} + 2OH^-\ (aq) \rightarrow \cancel{Na^+} + Cl^-(aq) + \cancel{Na^+} + ClO^-(aq) + H_2O\ (l)$
- Cl_2 and H_2O are both covalent so can't be split up into ions and are also not spectator ions

1c

c) The oxidation numbers of chlorine in each of the chlorine-containing ions formed are:

- OCl^- is +1
  AND
  Cl^- is -1; [1 mark]

[Total: 1 mark]

- The oxidation number of chlorine in OCl^- can be determined as follows:
  - Oxygen has an oxidation state of -2
  - (-2) + Cl = -1
  - Cl = +1
- The oxidation number of chlorine in the chloride ion can just be determined by looking at the charge on the ion

2a

a)

i) The equation is:

- Br_2 (g) + H_2 (g) → 2HBr (g); [1 mark]

ii) The trend in thermal stability of the hydrogen halides is:

Trend

- Decreases; [1 mark]

Explanation

- The atomic radius of the halogens increases down (the Group); [1 mark]
- Bond length increases; [1 mark]
- Less energy is required to break it / the bond; [1 mark]

[Total: 5 marks]

- The reactions between hydrogen and the halogens become less vigorous down the group
- The hydrogen halide formed is a gas
- As the atomic radius of the halogens increases down the group, the H-X bond has

to increase in length
- Longer bonds tend to be weaker bonds
- As the bonds get weaker, the hydrogen halogens become less stable to heat as you move down the group

2b

b) The order of the halogens from weakest to strongest oxidising agents is:

- Iodine < bromine < chlorine; [1 mark]

[Total: 1 mark]

- As you go down Group 17, the ability of the halogens to act as oxidising agents decreases
- Chlorine is a better oxidising agent than bromine because:
    - An oxidising agent will accept electrons from another substance
    - Going down the group, the atomic radii of the elements increase which means that the outer shells get further away from the nucleus
    - An 'incoming' electron will therefore experience more shielding from the attraction of the positive nuclear charge
    - The halogens' ability to accept an electron (their oxidising power) therefore decreases going down the group

2c

c) Iodide ions are better reducing agents because:

- An iodide ion / I^- is larger than a bromide ion / Br^-
  OR
  An iodide ion / I^- has a larger ionic radius than a bromide ion / Br^-; [1 mark]
- (Therefore the) outermost electrons are more shielded by inner shells; [1 mark]
- (Therefore the) outermost electrons are less strongly held / more easily lost; [1 mark]

[Total: 3 marks]

- A reducing agent donates electrons to other species
    - Therefore, it will be oxidised
- This question is asking you to compare two different ions, therefore you must

make sure your answer reflects this, for example:
  - An I^- ion is **larger** than a Br^- ion
  - If you do not say **larger** then you will not be awarded the mark

3a

a) The equation for the reaction between silver nitrate solution, $AgNO_3$ (aq), and sodium iodide solution, NaI (aq) is:

$AgNO_3$ (aq) + NaI (aq) → AgI (s) + $NaNO_3$ (aq)

- Correct formula for products; [1 mark]
- Correct balancing
  AND
  Correct state symbols; [1 mark]

[Total: 2 marks]

- When silver nitrate, $AgNO_3$, is added to a solution containing either chloride ions, bromide ions or iodide ions a precipitate is formed so the state symbol used must be (s)
  - Fluoride ions will not produce a precipitate as silver fluoride, AgF, is soluble in water
- The other product will be a solution containing the remaining ions which are Na^+ and NO_3^-
  - **Remember:** All nitrate salts are soluble in water
- You could also be asked to write the ionic equation for this reaction:
  - Ag^+ (aq) + I^- (aq) → AgI (s)

3b

b)

i) The yellow solid formed is:

- Yellow solid is sulfur; [1 mark]
- Smell of bad eggs is hydrogen sulfide; [1 mark]
- Purple vapour is iodine; [1 mark]

ii) The overall equation is:

$6HI (g) + H_2SO_4 (l) \rightarrow 3I_2 (g) + S (s) + 4H_2O (l)$

- Correct reactants and products; [1 mark]
- Corrects balancing; [1 mark]

[Total: 5 marks]

- For part (i)
  - Iodine is seen as a violet / purple vapour
  - The concentrated sulfuric acid oxidises hydrogen iodide and is itself reduced to sulfur
  - There is another product in this reaction which is sulfur dioxide gas
- For part (ii)
  - To write the equation:
    - $HI (g) + H_2SO_4 (l) \rightarrow I_2 (g) + S (s) + H_2O (l)$
  - Balance the oxygen atoms so there are four on each side
    - $HI (g) + H_2SO_4 (l) \rightarrow I_2 (g) + S (s) + \mathbf{4}H_2O (l)$
  - Balance the hydrogen atoms so there are eight on each side
    - $\mathbf{6}HI (g) + H_2SO_4 (l) \rightarrow I_2 (g) + S (s) + 4H_2O (l)$
  - Balance the iodine atoms so there are six on each side
    - $6HI (g) + H_2SO_4 (l) \rightarrow \mathbf{3}I_2 (g) + S (s) + 4H_2O (l)$

3c

c) The half equation is:

- $2Br^- \rightarrow Br_2 + 2e^-$; [1 mark]

[Total: 1 mark]

- Make sure your half equations are balanced for atoms **AND** charge
- You can write your half equation using the following steps
  - Write the formation of bromine gas from bromide ions
    - $Br^- \rightarrow Br_2$
  - Balance the bromines
    - $2Br^- \rightarrow Br_2 + e^-$
  - Balance the charge using electrons
    - $2Br^- \rightarrow Br_2 + 2e^-$
- When sodium bromide and sulfuric acid react, hydrogen bromide is produced initially

  - $NaBr(s) + H_2SO_4(aq) \rightarrow NaHSO_4(s) + HBr(g)$
- This is followed by the oxidation of HBr (g) and the sulfuric acid itself is reduced to sulfur dioxide gas
  - $2HBr(g) + H_2SO_4(aq) \rightarrow Br_2(g) + SO_2(g) + 2H_2O(l)$
- You could also be asked about the relative reducing power of the halide ions
- Iodide ions are stronger reducing agents than bromide ions so they can reduce the sulfur in sulfuric acid further
  - This is because
    - The outermost electrons experience more shielding by inner electrons
    - As a result of this, the outermost electrons are held less tightly to the positively charged nucleus
    - Therefore, the halide ions lose electrons more easily going down the group and their reducing power increases